EV Charging Reimbursement Report

User Guide & Logic Reference
Omexom NZ EV/PHEV/HYBRID Fleet

What this tool does

This report automatically fetches connected vehicle data from Compass IoT for the entire Omexom NZ EV fleet, detects every charging event, classifies each charge by location (Home / Office / Other), calculates reimbursement owed to drivers and costs to the company, and produces a per-driver analysis report you can use in one-on-one coaching sessions.

For BYD Shark PHEV vehicles, it additionally analyses the split between electric km and diesel km driven, and generates personalised recommendations on charging frequency to maximise EV utilisation.

1. Overview — What You Get

Running the script produces four output files in the reports\ folder:

File	Contents
EV_Charging_Report_<date>.csv	Raw data — every detected charging session across all vehicles and months. Suitable for import into Excel or Power BI.
EV_Charging_Report_<date>.xlsx	Formatted Excel workbook with three sheets: All Sessions (colour-coded), Summary by Vehicle, and Monthly Breakdown.
charging_map.html	Interactive Leaflet map. Every charging event plotted as a coloured pin. Click any pin to see driver, address, kWh, and zone.
driver_analysis_<date>.html	Interactive per-driver report. Select any vehicle from a dropdown to view their full charging profile, KPIs, insights, and (for PHEV) EV vs diesel km coaching panel.

1.1 The Driver Analysis Report

This is the primary tool for driver coaching sessions. Open driver analysis_<date>.html in any web browser — no internet connection required. Use the vehicle dropdown at the top to select a driver.

Each driver profile shows:

- **KPI strip:** total sessions, sessions by zone, total kWh, home reimbursement owed, company charging cost.
- **Insights panel:** auto-generated summary of charging behaviour, pattern classification, and flags for unusual activity.
- **Monthly breakdown table:** sessions and kWh split by HOME / OFFICE / OTHER per calendar month.
- **All charging sessions:** every event with start/end time, duration, SoC before/after, kWh, address, clickable Google Maps link, zone, and cost.
- **PHEV Analysis section:** (BYD Shark only) EV% vs diesel% bar, monthly km split, running cost breakdown, and driver coaching recommendations.

2. Data Source & How Data Is Fetched

2.1 Compass IoT Native Connect API

All vehicle data comes from the Compass IoT Native Connect REST API. This API provides connected vehicle telemetry directly from the onboard systems of BYD, Tesla, and Toyota vehicles — no additional hardware is required.

The script uses two API endpoints:

- **Historical Export (POINT):** Returns a time-series of GPS points for each vehicle, typically one point every 1–2 minutes while the vehicle is active. Each point includes: timestamp, latitude/longitude, speed, odometer (km), and FuelOrCharge (battery state of charge %). This is used for all vehicles to detect charging events.
- **Historical Export (TRIP):** Returns trip-level records for PHEV vehicles (BYD Shark). Each trip record includes a Path array of GPS points with per-point odometer and SoC values embedded as JSON. This higher-resolution data is used to split km into electric vs diesel modes.

> **Token:** The API requires a Bearer token. The script automatically obtains and caches this token using your workspace secret, refreshing it every 23 hours. The token cache is stored as .compass_token_cache.json next to the script.

> **Batch fetching:** Vehicles are sent to the API in batches of 50 VINs per request. If a batch returns a 400 error, the script automatically splits it in half and retries each half separately, down to individual VINs if necessary — isolating any unauthorised VINs without skipping valid ones.

2.2 Vehicle Coverage

The script covers 108 vehicles registered in the Compass IoT workspace, spanning:

- BYD Atto 3 — pure electric, 60.5 kWh battery
- BYD Seal — pure electric, 82.5 kWh battery
- BYD Shark PHEV — plug-in hybrid, 15 kWh electric-only battery, 100 km EV range
- Tesla Model Y — pure electric, 75 kWh battery
- Tesla Model 3 — pure electric, 57.5 kWh battery
- Tesla Model X — pure electric, 100 kWh battery
- Toyota RAV4 PHEV — plug-in hybrid, 18.1 kWh electric-only battery

Named drivers (24) have their name shown in reports. The remaining 84 vehicles are identified by their NZ plate number.

3. Charging Detection Logic

The script detects charging events by analysing the FuelOrCharge field (battery SoC %) across the time-series of GPS points for each vehicle. Two complementary detection methods run on every vehicle, and results are deduplicated.

3.1 Method 1 — Stationary SoC Gap (primary method)

This method is designed for BYD vehicles, which report continuously while parked.

- **Step 1 — Segment the timeline:** Points where speed < 1.0 km/h are grouped into stationary blocks. Moving segments are discarded.
- **Step 2 — Look for SoC increases:** Within each stationary block, the script looks for a sustained increase in SoC of at least 5 percentage points.
- **Step 3 — Validate duration:** The stationary block must last at least 30 minutes to qualify as a charging session. Sessions longer than 8 hours are excluded (vehicle is parked but not actively charging).
- **Step 4 — Calculate kWh:** kWh added = (SoC end % - SoC start %) / 100 × battery capacity (kWh) for that vehicle model.

3.2 Method 2 — Sleep-Gap Detection (Tesla supplement)

Tesla vehicles go offline when parked and stop reporting data. This creates gaps in the time-series that the stationary method cannot detect.

- **Identify gaps:** Any gap between consecutive data points greater than 60 minutes is considered a potential charging/sleep event.
- **Confirm charging:** If SoC is higher at the first point after the gap than at the last point before, a charging event is recorded. The minimum SoC increase threshold (5%) still applies.
- **Location:** The GPS coordinates of the first point after the gap (the wake-up location) are used as the charging location.

3.3 Charging Session Fields

Field	Description
Start / End Time	Timestamps of when charging began and ended (NZ local time).
Duration	Session length in minutes.
SoC Start / End %	Battery state of charge at beginning and end of session, rounded to 1 decimal place.
kWh Added	Energy added, calculated from SoC delta × battery capacity. Rounded to 2 decimal places.
Address	Street address from reverse geocoding (OpenStreetMap Nominatim). Falls back to GPS coordinates if geocoding fails.
Google Maps	Clickable link opening the exact charging coordinates in Google Maps for visual verification.
Zone	HOME, OFFICE, or OTHER — see Section 4.
Nearest Charger	Name of the closest known public charger to the charging location.
Home Reimb. (NZD)	Amount reimbursable to the driver if zone is HOME ($0.49/kWh).
Charging Cost (NZD)	Cost to the company if zone is OFFICE or OTHER ($0.32/kWh).

4. Zone Classification

Every charging session is classified into one of three zones based on its GPS position relative to Webfleet geofence areas.

Zone	What it means	Financial treatment
HOME	Vehicle charged at the driver's registered home address (Webfleet geofence areas 20000–20005, 20007–20015).	Reimbursable to driver at $0.49/kWh NZD.
OFFICE	Vehicle charged at an Omexom office or company charger (geofence areas 20006, 20017, 20018 — Silverdale, Mt Wellington).	Company cost at $0.32/kWh NZD.
OTHER	Vehicle charged at a public charger, roadside location, or any unregistered address. Verify against Google Maps link.	Company cost at $0.32/kWh NZD.

Note: If a driver's home address is not yet registered as a Webfleet geofence, their home charges will appear as OTHER. Contact the Direct-Track team to add the geofence. Each address in the All Sessions table has a clickable Google Maps link for verification.

5. Financial Calculations

5.1 Home Reimbursement

Rate: $0.49 per kWh (NZD)

Applied to all HOME zone sessions. This rate reflects the average residential electricity tariff in New Zealand and is the amount owed to the driver for charging at home.

Formula: Home Reimbursement = kWh Added × $0.49

5.2 Company Charging Cost

Rate: $0.32 per kWh (NZD)

Applied to OFFICE and OTHER zone sessions. This rate reflects the typical commercial electricity tariff and is the cost to the company for charges at company-owned or public facilities.

Formula: Charging Cost = kWh Added × $0.32

5.3 kWh Calculation

The amount of energy added is estimated from the change in battery state of charge (SoC) using the known battery capacity for each vehicle model:

Formula: kWh Added = (SoC End% - SoC Start%) ÷ 100 × Battery Capacity (kWh)

Vehicle Model	Battery Capacity Used
BYD Atto 3	60.5 kWh (usable)
BYD Seal	82.5 kWh (usable)
BYD Shark PHEV	15.0 kWh (EV-only usable battery)
Tesla Model Y	75.0 kWh
Tesla Model 3	57.5 kWh
Tesla Model X	100.0 kWh
Toyota RAV4 PHEV	18.1 kWh (EV-only usable battery)

6. PHEV Analysis — BYD Shark

The BYD Shark (plates RJW47 and RKU23) is a plug-in hybrid. In addition to standard charging detection, the script performs an EV vs diesel km split analysis to measure how effectively the vehicle's electric range is being used.

6.1 How EV vs Diesel km Is Determined

The Compass IoT TRIP endpoint returns a Path array for each trip. Each point in the path contains a GPS coordinate, an odometer reading (km), and a FuelOrCharge value (battery SoC %).

For each consecutive pair of path points, the script calculates:

- **Segment distance:** odometer[next] − odometer[current]
- **SoC drop rate:** (SoC[current] − SoC[next]) ÷ segment distance

If the SoC drops by 0.5% or more per km, the segment is classified as EV-powered (battery is being consumed by the motor). If SoC is flat or rising, the segment is classified as diesel-powered (engine running, battery maintained or recharged via regeneration).

> **Threshold:** SOC_DROP_THRESHOLD = 0.5% per km. Segments below this are diesel. This threshold accounts for minor SoC fluctuations during regen braking on diesel-dominated segments.

6.2 PHEV Cost Calculations

Metric	Formula / Value
EV running cost	EV km × 0.15 kWh/km × $0.32/kWh
Diesel running cost	Diesel km × 0.085 L/km × $2.20/L
Blended cost per km	Total running cost ÷ total km
Saving vs full diesel	Full diesel cost − full EV cost (potential saving if 100% of km on EV)
Charges needed/month	ceil(avg monthly km ÷ 100 km EV range)
Charges gap	Charges needed − current monthly charge frequency

6.3 EV Usage Rating

Rating	EV km %
Excellent 🌟	70% or above
Good 👍	45% – 69%
Needs Improvement ⚠	25% – 44%
Poor 🚨	Below 25%

7. Output Files in Detail

7.1 Excel Workbook (.xlsx)

Sheet 1 — All Sessions

Every detected charging session across all vehicles and months. Rows are colour-coded by zone: green (HOME), yellow (OFFICE), red (OTHER). Columns include all session fields listed in Section 3.3. The Google Maps column contains clickable hyperlinks labelled View Map.

Sheet 2 — Summary by Vehicle

One row per vehicle. Shows total sessions, home/office/other session counts and kWh, total home reimbursement, office cost, other cost, and total company cost. A grand totals row appears at the bottom.

Sheet 3 — Monthly Breakdown

Same columns as the Summary sheet but split by calendar month. Useful for month-on-month trend tracking.

7.2 Interactive Map (charging_map.html)

Open in any browser. Pins are colour-coded by zone. Click any pin to see a popup with the driver name, date, kWh added, SoC start/end, zone, and address. The map automatically fits to show all charging events.

7.3 Driver Analysis Report (driver_analysis_<date>.html)

Open in any browser. Use the dropdown at the top to select a vehicle. All data is embedded in the file — no internet connection or server required. The filter buttons below the sessions table allow filtering by HOME / OFFICE / OTHER to focus the conversation on specific charge types during a driver coaching session.

8. Running the Script

8.1 Requirements

- **Python 3.8+** with packages: requests, openpyxl, folium
- Install: `pip install requests openpyxl folium`
- **Script location:** Place ev_charging_report.py in your working folder alongside .compass_token_cache.json (auto-created on first run).

8.2 Running

```
cd C:\Users\FrancisWynne\Downloads\distribute-python-powershell-0.8.1\
python ev_charging_report.py
```

8.3 What Happens During a Run

- Webfleet geofence areas are loaded (home addresses and office zones). Existing known areas are reused; new ones trigger a Webfleet API call.
- All months are pre-fetched from Compass IoT in batches of 50 VINs. This is the most time-consuming step — 108 vehicles across 2 months = approximately 4–6 API calls.
- TRIP data is fetched separately for the 2 BYD Shark PHEV VINs for EV/diesel km analysis.
- Each vehicle is processed: charging events detected, addresses reverse-geocoded, zones classified.
- CSV, Excel, map, and driver analysis HTML files are written to the reports\ folder.

8.4 Configuration

Setting	Default / Notes
MONTHS_BACK	2 — number of calendar months to report on (January and February 2026 as of current run).
RATE_PER_KWH	$0.49 NZD — home reimbursement rate.
OFFICE_OTHER_RATE	$0.32 NZD — office and other charging cost rate.
DIESEL_COST_PER_L	$2.20 NZD — pump price used for PHEV diesel cost calculations.
MIN_CHARGE_PCT	5% — minimum SoC increase to qualify as a charging event.
MIN_DURATION_MINS	30 minutes — minimum duration of stationary period.
MAX_DURATION_MINS	480 minutes (8 hours) — maximum session cap to exclude long parked periods.
SLEEP_GAP_MINS	60 minutes — minimum data gap to trigger Tesla sleep-gap detection.
VIN_BATCH_SIZE	50 — maximum VINs per Compass IoT API request.

9. Interpreting Results in a Driver Coaching Session

9.1 Preparation

- Open driver_analysis_<date>.html in Chrome or Edge.
- Select the driver's vehicle from the dropdown.
- Have the Google Maps links ready — click View Map on any session you want to verify.

9.2 What to Cover

- **Total kWh and sessions:** Is the vehicle being charged regularly? Low session counts may indicate the driver is relying on public chargers or not charging at all.
- **Zone split:** High HOME% is expected and desirable — reimbursement is owed. High OTHER% warrants discussion about whether home charging is available and practical.
- **Address verification:** For HOME sessions, confirm the address matches the driver's registered home. Click View Map to verify. If a HOME charge appears at an unexpected address, the geofence may need updating.
- **OTHER charges:** Check the addresses. Public chargers (e.g. ChargeNet, Vector) are legitimate. Unusual locations should be discussed with the driver.
- **Session duration and kWh:** Very short sessions (< 1 kWh) may indicate the vehicle was plugged in briefly — not a full charge. Very long sessions may indicate the vehicle was left plugged in overnight at a public charger.

9.3 PHEV Coaching (BYD Shark)

- **EV% bar:** The target is 70%+. If the bar shows below 30%, the driver is predominantly running on diesel and not charging frequently enough.
- **Charges needed vs current:** The report states how many charges per month are needed to cover all km on electric, and the gap versus current behaviour. Use this as a concrete, actionable target.
- **Cost saving:** Show the driver the 'Saving vs Full Diesel' figure. This is the financial benefit realised by charging the vehicle before each working day, expressed in dollars.
- **Practical advice:** Charge at home overnight before a working day. Plug in at the office when available. The Shark has a 100 km EV range — most NZ daily commutes and site visits fall within this range.

10. Known Limitations & Accuracy Notes

- **SoC rounding:** Compass IoT reports SoC in whole percentage points. For a BYD Atto 3 (60.5 kWh), 1% SoC = ~0.6 kWh, so kWh figures are accurate to approximately ±0.3 kWh per session.
- **PHEV EV/diesel split:** The 0.5%/km threshold is a calibrated estimate. Motorway driving at high speed with regenerative braking can cause minor SoC fluctuations that may slightly misclassify short segments. Overall trip-level accuracy is expected to be within ±5%.
- **Geocoding speed:** Reverse geocoding uses OpenStreetMap Nominatim, which is rate-limited to one request per second. A full run with many unique locations can take several minutes. Addresses are cached between runs.
- **Fleet Partners Teslas:** 47 Tesla vehicles registered under fleet.partners.co.nz are visible in GetVehicles but are not authorised in the Omexom Compass workspace. They return no data and are excluded from the report.
- **Toyota RAV4 PHEV:** VIN JTMAAABJ004070717 (labelled COMPASS TOYOTA) uses 18.1 kWh as the battery capacity. The actual usable capacity may differ — verify with the vehicle's documentation.
- **Data gaps:** If a vehicle has no data for a month (e.g. it was off-fleet, in for service, or the telematics were inactive), it will show 'No data' in the console and contribute zero sessions to the report.